Shreya Kalyanaraman

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Portfolio
Linkedin

EDUCATION

University of Maryland — AUG 2024 - MAY 2026
M.Eng. in Robotics — COLLEGE PARK, MD

TECHNICAL SKILLS

Programming & Development: Python, C++, ROS2, Git, Linux, Embedded C
Machine Learning & AI: PyTorch, JAX, TensorFlow, Reinforcement Learning (PPO, TD3), Imitation Learning, NumPy, SciPy, scikit-learn, OpenAI Gym
Motion Planning & Control: A*, Dijkstra, RRT, RRT*, PRM, Control Systems (PID, LQR/LQG), MoveIt
Robotics Hardware: Universal Robots (UR3e, UR5e, UR10e), Franka Emika Panda, Turtlebot 4
Simulation : Gazebo, RViz, PyBullet, MuJoCo, CARLA, SAPIEN, NVIDIA Isaac Sim

EXPERIENCE

Artly AI – Belta Research Group — JAN 2026 - PRESENT
Robot Learning Intern — COLLEGE PARK, MD

- Built VR teleoperation stack on xArm7 and Z-Arm + RealHand L6 using DimensionalOS, and developed MuJoCo simulation environments for dexterous manipulation research.
- Deployed ICP-based perception pipeline with FPFH global registration on FR3 + RealHand L6, achieving 100% grasp success across 24 hardware trials on 6 YCB objects.

RAAS Lab - Maryland Robotics Center — JAN 2025 - PRESENT
Research Assistant — COLLEGE PARK, MD

- Developed a transfer learning framework that infers action semantics at inference time, enabling single RL agents to operate across heterogeneous state-action spaces, tested on OpenAI Gym Retro.
- Deployed and fine-tuned Pi-Zero policies on a UR3e for table bussing tasks.
- Calibrated and integrated a multi-camera perception system with three Intel RealSense depth cameras to enable multimodal feedback for manipulation tasks.

IIT Madras - Guidance, Navigation, & Control Lab — FEB 2024 - JUL 2024
Research Intern — CHENNAI, INDIA

- Implemented the Gradient Direction Turn Switching (GDTS) algorithm in ROS 2/Gazebo for UAV search-and-rescue source localization, improving trajectory efficiency in dynamic environments.

PROJECTS

PPO and DDPG Based Parallel Parking Agent in CARLA link

- Developed and benchmarked deep reinforcement learning agents (PPO/DDPG) on time-series sensor data for autonomous parking; achieved 93% + success rates in dynamic scenarios using 500+ trial dataset.
- Designed real-time sensor fusion and neural-policy pipelines leveraging LiDAR, radar, and visual streams; applied curriculum learning and reward shaping to reduce maneuver time by 20% while maintaining 95% safety compliance.

Deep Multi-Agent Highway On-Ramp Merging in Mixed Traffic link

- Reimplemented and extended the Deep Multi-Agent Reinforcement Learning for Highway On-Ramp Merging framework (Chen et al., arXiv:2105.05701) in OpenAI Gym using PPO; integrated action masking and prioritized safety supervision to boost collision reduction from 20% to 30% and merge speed gains from 10% to 15%.
- Employed centralized training with decentralized execution and curriculum learning across increasing traffic densities, sustaining greater than 90% merge success under high-volume scenarios.

Autonomous Barista Robot link

- Led team of 4 in developing a ROS2 package for a 6-DOF UR10e manipulator, simulating coffee-making processes in a custom Gazebo environment with RViz visualization.
- Designed position, velocity, and effort controllers that achieved precision in end-effector path following; reduced trajectory deviation to less than 2mm during operation.

Lerobot Hackathon 2025

- Assembled dual-arm Koch manipulators and developed Python control wrapper for synchronized joint-state feedback, collecting demonstration data via active-passive mimicking for pick-and-place tasks.
- Trained Behavior Cloning and Action Chunking Transformer (ACT) policies on the Koch arm, achieving a 92% task success rate.